

CoordAuth: Hands-Free Two-Factor Authentication in Virtual Reality Leveraging Head-Eye Coordination

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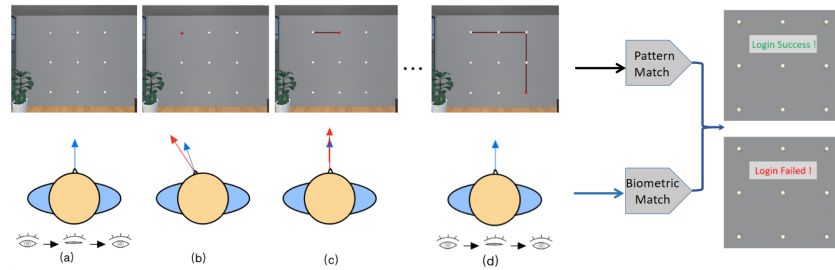


Figure 1: A storyboard depicting an authentication attempt for the pattern “12369” with CoordAuth. (a) The interface features a 3×3 grid pattern with a 60° grid size. The user blinks and reopens their eyes over a duration of 0.3s to start the authentication attempt. (b) To input “1”, the user rotates the head and eyes to focus on the hit test region (as Figure 2(a) shown). The blue arrow indicates the head’s facing direction, while the red arrow represents the eye gaze direction. (c) Then user rotates their head and eyes to gaze across the hit test region at the center point of the first row to input “2” (as Figure 2(b) shown). (d) Once the user completes the pattern, they blink and re-open their eyes to conclude the authentication. (e) CoordAuth evaluates the login attempt by considering both the match of the input pattern and the behavioral biometrics derived from head-eye coordination movements.

ABSTRACT

We present CoordAuth, a gaze-based two-factor authentication technique in VR utilizing implicit head-eye motion features to offer a more secure and natural alternative to traditional pattern-based authentication. Users authenticate by performing gestures across 3×3 grid points using their eyes. We first optimized CoordAuth’s UI by evaluating participants’ input performance and experiences across different grid sizes. Then we extracted the head-eye coordination features during pattern entering to construct CoordAuth’s authentication algorithm, which ensembles Random Forest classifiers across saccade and fixations segments. CoordAuth demonstrated strong security with a 1.6% FAR and 1.5% FRR across 24 registered users in the password collision scenarios. A subsequent study demonstrated that CoordAuth achieved an 0.6% Attack Success Rate (ASR) against shoulder-surfing attack from a 1-meter distance. Usability evaluations in standing, sitting, and moving postures showed that CoordAuth achieved significantly faster authentication speed and lower rejection rate than laser and touch-based baselines. Meanwhile, it was the most preferred by the participants in terms of social acceptance and physical effort.

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Index Terms: Authentication, Behavioral biometrics, Gaze interaction, Fixation and saccade, Virtual Reality, Applied Machine Learning

1 INTRODUCTION

As Virtual Reality (VR) technology expands across domains like gaming [5], e-commerce [39], and networking [48], the need for secure, user-friendly, and privacy-conscious authentication is growing [23]. This is especially important as VR devices are increasingly shared among users, including families, coworkers, and students [59]. Strong security measures are essential to ensure safe access, secure applications, and the protection of sensitive operations [56].

In VR, the hands-free interaction paradigm is highly desirable as it enhances immersion, reduces the need for additional hardware [23], and allows for a more natural and seamless user experience [67]. Traditional input devices, such as keyboards or controllers, can break the sense of immersion and limit the fluidity of interaction within the virtual environment [55]. Therefore, authentication methods that can operate without the need for extra physical input devices are not only more convenient but also align better with the core principles of VR design [18].

However, achieving secure, efficient, and accurate hands-free authentication presents significant challenges. Existing authentication methods, including knowledge-based (such as visual passwords [56, 13, 15], PINs [66, 16] and patterns [16]), possession-based (utilizing biological features like ears [14] and irises), often fall short under these constraints. Knowledge-based approaches are vulnerable to shoulder-surfing and require the use of hand, while possession-based methods raise privacy concerns and often require additional hardware [56].

Given these limitations, ensuring both robust security and seam-

less user experiences demands innovative solutions that address the vulnerabilities and usability challenges of existing approaches. Behavioral biometrics, such as head and eye movements [28], full-body motion [59], and hand gestures [57], offer potential solutions but also come with their own challenges. These methods often suffer from higher error rates and longer authentication times [42], or they require additional input devices, such as controllers [57], which can conflict with the goal of achieving seamless, hands-free interaction. To reduce the high error rates and enhance security, recent advancements explored two-factor authentication (2FA), which further combined behavioral biometrics with knowledge-based methods [67, 63]. However, existing 2FA methods often employ uncommon interactions, such as eye blink rhythms [67], which can be cumbersome and difficult to learn and remember. Or some methods rely on controllers [31], which conflicts with the goal of achieving seamless and efficient hands-free authentication.

To address the previous challenges, we proposed CoordAuth, a hands-free 2FA method for VR environments. Through entering the 3×3 pattern password by gaze in VR, CoordAuth assesses the correctness of pattern as the first authentication factor (knowledge-based). Additionally, the implicit biometric features extracted from head and eyes movements construct the second authentication factor.

As the motion capability of eyes and heads were significantly different from fingers, we first conducted a study to investigate how grid size impact user's input behaviours, in the mean time set UI design factors. Through setting 4 grid sizes (40°-100° Field of View, FOV), we concluded 60° was the most suitable regarding input speed, error rate and subjective ratings. We observed that different users exhibited distinct characteristics of head movements, which inspired the design of our authentication algorithm.

The process of gaze input for a pattern can be naturally divided into saccades and fixations [1, 36]. Building on this, CoordAuth utilizes head and eye movement features (e.g., velocity, frequency spectrum) and different segments (i.e., saccades and fixations) for behavioral biometric authentication. We collected 24 participants' data for simulation with 20 selected patterns and identified Random Forest as the most accurate classifiers among candidates. A voting mechanism combines different types of classifiers (i.e., fixation and saccade classifiers) for authentication, achieving an False Rejection Rate (FRR) of 1.5% and an False Acceptance Rate (FAR) of 1.6% in the extreme case of password collision. A 7-day longitudinal simulation with 10 users confirmed the effectiveness and robustness of these behavioral biometric features. A subsequent study assessed CoordAuth's security, demonstrating its resilience against shoulder-surfing attacks with Attack Success Rate (ASR) of only 0.6% at even 1 meter.

In Study 2, We evaluated CoordAuth's usability against commercial device authentication methods: PIN using touch-based interaction and pattern using laser-pointing interaction. Besides, we also compared CoordAuth to gaze-based pattern authentication [56]. CoordAuth achieved faster login speed (4.6s) compared to Touch (6.3s) and Laser (5.3s) in sitting postures. Additionally, CoordAuth (3.2%) outperformed Touch PIN (17.4%) and Laser Pattern (15.8%) in terms of FRR. Similar trends were also observed in standing and walking postures, demonstrating the robustness of CoordAuth. CoordAuth maintained comparable authentication time (4.6s v.s. 5.6s) and comparable FRR (3.2% v.s. 6.3%) despite incorporating an additional authentication factor, when compared to the gaze-only baseline. Subjective feedback highlighted CoordAuth's enhanced generalizability, novelty and social acceptability.

The contributions of this paper are threefold:

- We analyzed the effect of grid size (40-100° FOV) on user input behavior while entering 3×3 grid patterns using gaze in VR, finding that a grid size of 60° yielded the best performance.

- We introduced CoordAuth, a natural hands-free 2FA technique in VR leveraging head-eye coordination motion features, employing voting mechanisms to enhance security.
- We systematically evaluated CoordAuth's security and usability through two user studies, demonstrating comparable authentication speed with reduced error rates relative to hand-based methods, ensuring security against shoulder surfing attack.

2 RELATED WORK

2.1 Hands-free Interaction Leveraging Head and Eye Movements

Eye movement is a natural hands-free interaction method, where individuals instinctively gaze at target objects [30]. However, unintentional gaze interactions can lead to the Midas Touch problem [22]. To address this, strategies like dwell time [36], simulated mouse clicks [7], and optimized hit test regions [17, 36] have been developed. Meanwhile, hands-free interaction methods based on head movements, such as HeadGesture [61] and intention detection [60], have also been explored.

Integrating eye movement with head movement creates a more seamless and effective interaction method [36], without using hand controls [44]. This approach enhances performance in tasks like object selection [10], aids in correcting calibration drift [44], and boosts overall interaction efficiency [24], especially for text entry tasks. Research comparing gaze and head movement inputs in Augmented Reality (AR) and Virtual Reality (VR) environments [24, 43] indicates that gaze-based input techniques, where the head is free to move, are faster and more precise [43, 24], whereas head-only methods may lead to greater fatigue [43]. However, these findings have not yet been applied in authentication contexts, where they hold significant potential to make authentication more natural and efficient.

2.2 Head-eye Coordination Modeling

As a natural and efficient interaction method, head-eye coordination has been widely studied in both biological field [11] and Human Computer Interaction (HCI), particularly in VR. It involves the synchronized movements of the eyes and head during gaze shifts [52]. This coordination is essential for expanding the field of view (FOV) beyond the 50° range of typical eye movement [25], allowing for a broader exploration of virtual environments. During rapid gaze shifts, the head and eyes move together to reduce eye strain, with the head continuing to move after the gaze target is reached, while the eyes make compensatory movements [58].

Smaller gaze shifts under 20° are typically managed by the eyes alone [25, 11], while larger shifts require head movement [49]. Factors influencing head movement include initial eye position, target amplitude, and location [38, 12]. Individual variations, such as the tendency to rely on head movements, lead to distinctions like "head movers" and "non-head movers" [12]. These differences are critical in understanding user behavior in VR tasks, such as entering pattern passwords within specific FOVs, forming the basis for feature selection in behavior-based biometric systems like CoordAuth.

2.3 VR Authentication Techniques Leveraging Behavioral Biometrics

Research in the VR domain focuses on developing novel and VR-specific authentication technologies that fall into five major categories [56, 2]: knowledge-based [32, 15], physical biometric [50, 14, 68], behavioral biometric [53, 37, 46, 29, 34], token-based [4], and multi-factor [63, 31, 67].

Among these, behavior biometrics, which leverages human's motion control ability to form authentication patterns or factors, are particularly notable for their lower privacy risks and the resistance

against shoulder surfing [20]. There is a considerable body of research on various aspects of behavior biometrics, including head movement [45, 42], eye movement [37], hand gestures [41, 37], blink patterns [47, 67], and full-body motion [33, 37, 29, 27]. While some studies integrate head and eye movements, it is more common to focus on single features [45, 37]. Although some previous works [37, 47] simultaneously employed both, they did not delve into the characteristics of head-eye coordination movements. Additionally, Current behavioral biometric authentication methods often rely on machine learning or proprietary models with limited interpretability [35]. In contrast, our head-eye coordination model offers both theoretical support and explanatory insights for behavioral biometric features in VR.

Behavioral biometrics are often used to enhance the security of two-factor authentication (2FA) systems, employing unique physical traits (such as BlinkEye's blink patterns [67], RubicBiom's hand movements [31], and GlassGesture's head movements [63]) as additional authentication references. Inspired by this approach, CoordAuth leverages behavioral biometrics to improve the security and trustworthiness of authentication.

3 INTERACTION DESIGN AND IMPLEMENTATION OF COORDAUTH

3.1 UI Design

As shown in Figure 1, our design utilizes the comfortable gaze interaction for entering patterns in VR, adopting a 3×3 grid pattern lock similar to smartphone interfaces. In the VR setting, users interact with a vertically aligned 3×3 grid, where each grid point is initially a white sphere that turns red upon gaze to signal selection. A red line connects sequentially selected points, and each point can only be selected once per entry. Selecting endpoints of a straight line automatically selects all intervening points.

Our pilot study identified the optimal size of the grid points, balancing ease of targeting with the potential psychological impact of oversized objects in VR [17]. Based on this study and existing literature [36], we determined the point size as a 3° FOV.

3.2 Interaction Design

Our design minimizes user interaction complexity in VR by utilizing non-spontaneous eye blinks to initiate and terminate input, without requiring hand movements. We define one complete closure of the upper and lower eyelids as one blink. In our pilot study, participants were instructed to intentionally close their eyes to initiate the input. We found that such blinks typically lasted over 0.3 seconds, with some extending beyond 1 second. Previous research also indicated that spontaneous eye blinks in humans last approximately 0.1-0.15 seconds [3]. Therefore, we empirically set a criterion that a non-spontaneous blink lasting at least 0.3 seconds triggers the start and end of input, effectively preventing accidental activations caused by spontaneous blinking.

3.3 Hit Test Detection

To alleviate the "Midas Touch" problem prevalent in eye gaze interactions [22, 17], we adopted the goal-crossing strategy [17] with an enlarged hit-test area for object selection in VR. This approach allows for selection by merely having the gaze point intersect the hit-test area for a single frame.

Our pilot study assessed hit test regions of $\pm 4^\circ$, $\pm 5.5^\circ$, $\pm 7^\circ$ and $\pm 8.5^\circ$ FOV around the object's center, finding that a $\pm 5.5^\circ$ FOV minimized error rates effectively. While this dimension was larger compared to some previous findings [36, 52], it was justified given the typical gaze drift in eye-tracking devices [9]. The specific parameters and interaction details are illustrated in Figure 2.



Figure 2: (a) Parameters for the hit test region. (b) Selection of targets based on goal crossing.

4 STUDY 1: EXAMINING THE EFFECT OF GRID SIZE ON USERS' INPUT BEHAVIOUR

Previous literature found the grid size has significant influence on gaze input speed and error rate performance [12]. Thus, we conducted a user experiment to determine the optimal size for the UI.

4.1 Participants and Apparatus

We recruited 16 participants from the campus (7 females, 9 males, average age = 23.8, SD = 2.1), all with normal or corrected vision. The average self reported VR usage experience was 2.6 (minimum 1, maximum 5, SD = 0.56). Two participants had prior experience with VR authentication. Each participant was compensated 15\$ for their participation. The study was reviewed and approved by the Institutional Review Board (IRB) of our institution. The study used a Meta Quest Pro headset. We utilized the Oculus integration SDK for high-frequency eye tracking¹. The API reported the eye tracking and head movement data both at 50Hz. The experimental platform was developed in C# using Unity 3D 2021.3.21f1.

4.2 Study Design

We employed a **within-subject** design, with **grid size** as the only factor. Notably, head movements begin to emerge when the FOV spans 40° - 50° [25, 11], prompting head-eye coordinated movements. To optimally exploit head-eye coordination features, we set 40° FOV as the minimum grid size. In alignment with related research [52] that adjusts head-eye coordination for gaze selection in 20° FOV increments, we selected four grid sizes: 40° , 60° , 80° , and 100° , as shown in Figure 3.

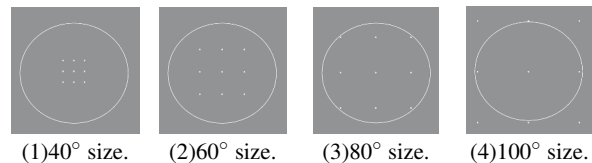


Figure 3: Grid layouts of different sizes, with the white ellipse representing the FOV of the head-mounted display.

To simulate authentic pattern entry, we instructed participants to input 20 pre-designed patterns [6] with different grid sizes. These patterns are among the most commonly used in Android, so we deemed them sufficiently representative. We asked participants to input these patterns as quickly and accurately as possible.

4.3 Procedure

We initially introduced participants to the experiment platform and usage of CoordAuth, allowing them five minutes for practice. Then the main experiment was divided into four sessions differed by the grid size (see Section 4.2) with a counter-balanced order through Latin Square design. For each session, participants would undergo

¹<https://developer.oculus.com/downloads/package/unity-integration/>

20 patterns' instruction and input in a random order using the shuffle function of Python. For each pattern, we first showed the drawing of the pattern using green lines and points for one time. After the showing disappeared, users repeated the pattern by themselves. The pattern drawn was rendered in red. We did not inform participants whether their pattern was correct. Participants filled in the questionnaire after each session.

4.4 Results

We collected 1280 trials across all participants (16 participants $\times$ 4 sizes $\times$ 20 PINs). The collected data were assessed for normal distribution using the Shapiro-Wilk normality test. For data deviating from normal distribution, we employed the non parametric tests method Friedman test. For data that followed a normal distribution, repeated measures (RM) ANOVA was employed. All post hoc pairwise comparisons were protected against Type I error using the Bonferroni correction. The statistical significance level was set at .05 for all analyses

4.4.1 Input time

The input time refers to the average duration required for users to draw each pattern, including the non-spontaneous blinks that indicate the start and end of the input process. Through observation of the experiment and the data, we removed unreasonable input time outliers ($< 2s$ or $> 10s$) caused by trigger detection errors, which accounted for 0.9% of the total data.

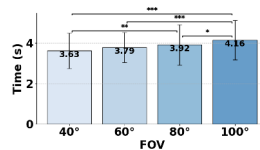


Figure 4: Bar plots showing main effects of size on the input time. The error bar shows one standard error.

Figure 4 illustrates the averaged input time across each size. RM-ANOVA revealed a statistically significant main effect of size on input time ($F_{3,45} = 16.5$, $p < .001$). Post-hoc results revealed significant differences between 40° and 80° ($p < .01$), 40° and 100° ($p < .001$), 60° and 100° ($p < .001$), as well as 80° and 100° ($p < .05$). These results indicated that larger size than 60° would increase input time significantly. **To sum up, in terms of input time, 40° and 60° outperformed 80° and 100°.**

4.4.2 Input Error Rate

We calculated input error rate with Exact Matching (total) and Pattern Matching (digit-wise) methods. Exact Matching refers to scoring 1 if the input pattern and given pattern are identical and 0 if they are not the same. Pattern Matching refers to the proportion of incorrectly input digits. Figure 5 illustrates the average input error rate across four sizes.

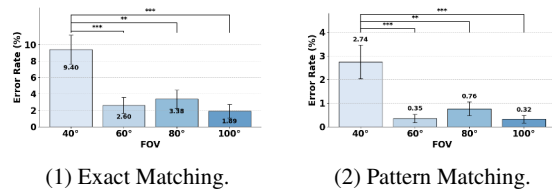


Figure 5: Bar plots showing average input error rate calculated by (1) Exact Matching and (2) Pattern Matching. The error bar shows one standard error.

RM-ANOVA revealed a significant difference in input error rate using both Exact Matching ($F_{3,45} = 7.77$, $p < .001$) and Pattern Matching ($F_{3,45} = 8.11$, $p < .001$) strategies. Post-hoc comparisons for Exact Matching showed significant differences between 40° and 60° ($p < .001$), 80° ($p < .01$), and 100° ($p < .001$). Similarly, Pattern Matching revealed significant differences between 40° and other sizes with p-values of $p < .001$, $p < .01$, and $p < .001$, respectively. The results indicate that the **40° size is the most error-prone**, which may be due to error inputs caused by the smaller input grid size while the hit test region remains unchanged. This aligns with previous findings that dense layouts may cause users to make more errors [54].

4.4.3 Subjective Ratings

We collected participants' subjective ratings for different sizes using a 7-point likert scale questionnaire. The dimensions included perceived speed [64], perceived accuracy [64], and 5 dimensions from NASA-TLX. Figure 6 shows the scores across the different sizes for each dimension.

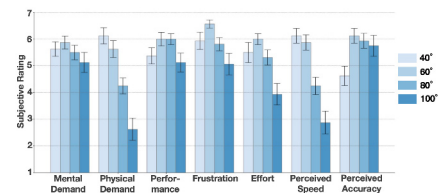


Figure 6: Subjective ratings across different dimensions (7: most positive, 1: most negative). The error bars indicate the standard error.

Among the four sizes, the Friedman test revealed significant differences in both *physical demand* ($p < .001$) and *perceived speed* ($p < .001$). Post-hoc analysis showed that for these two items, subjective ratings decreased monotonically with increasing size. No significant difference was found between 40° and 60°, but all other size comparisons revealed significant differences. This aligns with previous findings that larger sizes are linked to higher physical demand [54]. In conclusion, **40° and 60° groups could achieve comparable physical demand and perceived speed, while outperforming other grid sizes.**

4.4.4 Input Size determination and Head-Eye Coordination Analysis

Considering the input time (where 40° and 60° are better), input error rate (where 40° is the worst), and subjective ratings (where 40° and 60° are better), **we chose the 60° size.**

As described in Section 2.2, gazing at a target involves both head and eye movements, a process referred to **head-eye coordination**. In this study, when participants gaze at specific points located at the four corners and the four edge points of a 3×3 grid, the overall rotational contribution is a combination of head movement (HM) and eye movement (EM) relative to the head. Studying HM or EM is equivalent. For ease of analysis we focus on HM in this study. We observed a distinction between "head movers," where HM plays a predominant role, and "non-head movers," where EM is the primary contributor. This distinction aligns with findings from previous research [12, 52].

Figure 7 illustrates the probability density distribution of HM across every participants at all timestamps for gazing targets at the same distance.

For example, P10 shows a distribution concentrated in regions of larger HM, which means it is more probable to move its head at larger scale, identifying it as a "head mover". In contrast, P12's distribution is primarily within regions of smaller HM, categorizing

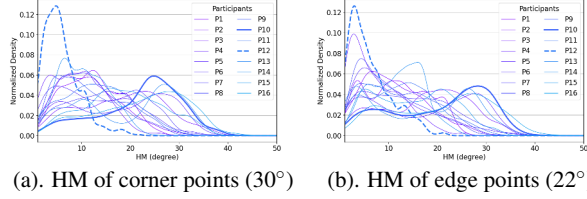


Figure 7: Probability density distribution of head movement (HM) for participants at target distances of 30° (a) and 22° (b). The example differences are shown between P10 ("head mover") and P12 ("non-head mover").

it as a "non-head mover". This variation illustrates that different individuals exhibit varying propensities for **head-eye coordination**. These observations inspired the development of our authentication algorithm in Section 5.

5 TWO FACTOR AUTHENTICATION ALGORITHM OF COORDAUTH

5.1 Algorithm Flow

Similar to most authentication methods [65], CoordAuth consists of two stages: registration and login. In the registration phase, users are required to input their patterns multiple times. Throughout this process, CoordAuth's Inertial Measurement Units (IMUs) and eye trackers gather head and eye movement data for behavioral biometrics. The data collection from these two devices is synchronized, with a sampling rate of 50 Hz. Simultaneously, the device records the patterns inputted by the user as a knowledge-based password.

During the login process, users re-enter their pattern password, CoordAuth's two-factor authentication mode verifies both the pattern and the head-eye movement features to confirm identity. Figure 8 shows CoordAuth's two-factor authentication algorithm. Since comparing pattern passwords is straightforward, our focus here is on exploring the behavioral biometric authentication.

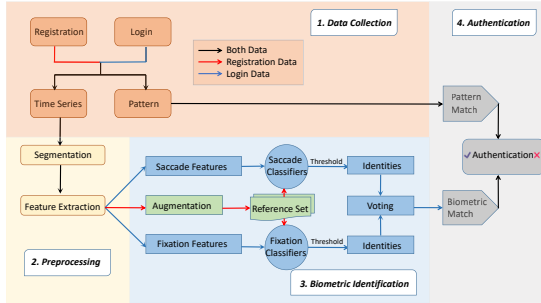


Figure 8: CoordAuth's authentication algorithm flow.

5.2 Segmentation of Head and Eye Sequences

Leveraging previous studies [1, 36] that indicate the interval between two saccades typically ranges from 100 to 200 milliseconds (ms), we defined our segmentation criterion: If a user's gaze remains within a point's hit test region for over 200 ms, the time sequence from when the gaze enters the hit test region to when it exits is splitted and categorized as a *fixation segment*. Consequently, the entire time sequence is segmented accordingly, with segments not classified as fixation being defined as *saccade segments*. Furthermore, each fixation segment trains the corresponding *fixation classifier* and each saccade segment trains the corresponding *saccade classifier*.

5.3 Feature Extraction

To minimize the impact of the user's initial position and posture, We collected initial data about the rotation and position of the head and eyes when the user is facing the 3x3 unlock UI. Based on this, we calculated the position and rotation of the head and eyes in the coordinate system of the initial posture.

The raw data from both the IMU and eye tracker represent position as a three-dimensional vector, $[x, y, z]^T$, and rotation as a four-dimensional quaternion, $[q_w, q_x, q_y, q_z]^T$, which corresponds to a specific rotation matrix [59]:

$$\mathbf{R}(\mathbf{q}) = \begin{bmatrix} 1 - 2q_y^2 - 2q_z^2 & 2q_xq_y + 2q_zq_w & 2q_xq_z - 2q_yq_w \\ 2q_xq_y - 2q_zq_w & 1 - 2q_x^2 - 2q_z^2 & 2q_yq_z + 2q_xq_w \\ 2q_xq_z + 2q_yq_w & 2q_yq_z - 2q_xq_w & 1 - 2q_x^2 - 2q_y^2 \end{bmatrix}$$

The Turning (Yaw), Nodding (Pitch) and Tilting (Roll) could separately be represented as $\text{Yaw} = \text{atan2}(\mathbf{R}_{21}, \mathbf{R}_{11})$, $\text{Pitch} = -\text{asin}(\mathbf{R}_{31})$, $\text{Roll} = \text{atan2}(\mathbf{R}_{32}, \mathbf{R}_{33})$, where $\mathbf{R}_{ij}$ represents the element in the i -th row and j -th column of the matrix $\mathbf{R}(\mathbf{q})$.

For head movement over time, there are **six** corresponding time series: $x(t)$, $y(t)$, $z(t)$, $\text{Yaw}(t)$, $\text{Pitch}(t)$, and $\text{Roll}(t)$. Regarding eye movement, the position vector has a fixed offset relative to the head, meaning there are no independent degrees of freedom. Consequently, the eyes generate only **three** time series with independent degrees of freedom: $\text{Yaw}(t)$, $\text{Pitch}(t)$, and $\text{Roll}(t)$.

In Section 4.4.4, we explored individual differences in behavioral biometric differences among individuals. These differences are crucial for our model's feature extraction. Drawing on prior research [67, 53] and biometric signal processing methods [65], we identified the following features:

- **F1. Raw:** The following statistical features: maximum, minimum, mean, median, and variance, are extracted for each raw time series, resulting in five dimensions.
- **F2. Deviation:** We computed the first derivative with respect to time for each time series (representing linear or angular velocity in physics). We then extracted the same six statistical features from these derivative time series as those in F1. Consequently, each time series in F2 has five dimensions.
- **F3. Spectrum:** For each time series, we applied the Fourier Transform (FT) to convert it into the frequency domain and then extracted four features: spectral energy, spectral centroid, spectral bandwidth, and peak frequency position. Thus, the F3 features for each time series have four dimensions.

In summary, the overall feature dimension for a single time series is $5 + 5 + 4 = 14$. As discussed above, there are 9 time series with independent degrees of freedom for head and eye movements. Consequently, when all features are included, a single saccade or fixation segment encompasses a total of $14 \times 9 = 126$ dimensions. Figure 9 illustrates the details of feature extraction.

We removed outliers from the original data and applied data augmentation techniques, adding Gaussian noise to the original time series, to enhance the training set. The augmentation ratio is empirically set as 20, aligning previous literature [65, 26]. Each feature in the training set is normalized to have a mean of 0 and a standard deviation of 1 / 3 [65]. The test set is normalized using the same mean and standard deviation as the training set.

5.4 Model Selection

After feature extraction, we employed classification models to implement the authentication. We utilized six leading and commonly used classification methods to avoid incurring excessive training computational costs [67, 31, 27]: Random Forest (RF), Support Vector Machine (SVM), K-Nearest Neighbors (KNN), Multilayer Perceptron (MLP), eXtreme Gradient Boosting (XGBoost), and Long Short-Term Memory (LSTM). We partitioned the training set and test set based on data collected to determine the optimal method. Using the scikit-learn, xgboost, and tensorflow libraries in

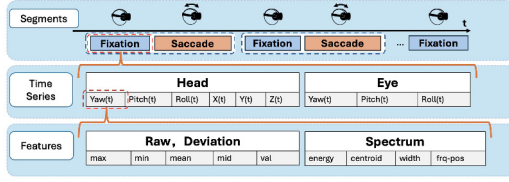


Figure 9: Illustration of feature extraction.

Python [40]. For LSTM, we split cross-validation set to roughly determine the optimal parameters for comparison. Other classifiers were implemented with default hyperparameters for reproducibility and fair comparison, as previous researches adopted [59].

We recruited 24 participants (13 males, 11 females; mean age = 25.3, SD = 1.6) from our campus, none of whom had participated in prior studies. The average self-reported VR experience was 2.9 (SD = 0.6) on a 5-point scale. Each participant received \$15 compensation. We used the Meta Quest Pro Headset and Oculus Quest Link to implement the authentication system. 20 patterns ranging from simple to complex were selected from previous works, with evenly distributed complexity scores [62, 6]. These patterns were designed with diverse starting points, numbers of intersections, and lengths, as shown in Figure 10. Each pattern was entered 10 times by each participant, yielding 4800 samples (24 participants $\times$ 20 patterns $\times$ 10 entries). To evaluate the performance of the saccade and fixation classifiers under different training sizes, T samples were used for training and the remaining 10-T samples for testing.

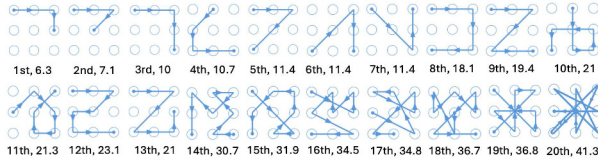
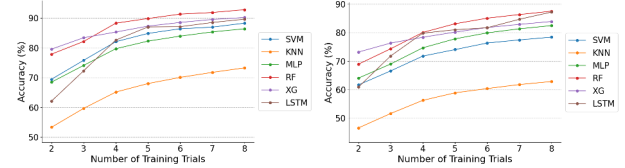


Figure 10: 20 patterns and their complexity scores. Pattern index and score are separated by a comma. $Complexity\ score = S \times \log_2(L + I + O)$, where S are the number of connected dots, L are the total length of all line segments, I are the number of intersections and O are the number of overlapping line segments [62].

Figure 12 presented the classification accuracy results, Random Forest consistently outperformed other methods, across most training sizes (from T = 4 to 8) for both classifiers. At T = 8, the saccade classifier's accuracy reached 92.8%, and the fixation classifier's accuracy reached 87.5%. We chose **Random Forest** as our final model because of its strong performance and its effectiveness in handling complex features. Additionally, we found that as the number of training trials increased, the accuracy did not plateau after reaching a certain number of trials. This inspired us to ensemble these classifiers using a specific method. Before exploring this, we noted that the accuracy for fewer than 5 attempts was significantly lower, while the accuracy for 5 attempts was significantly higher. Therefore, we ultimately set the training trials (i.e. **registration times**) to 5.

5.5 Ablation Analysis of Feature Sets

We then conducted an ablation analysis to determine the relative importance of all the extracted features for both classifiers. Overall, there were two types of ablation pathways: 1) ablating certain features from F1, F2, or F3; and 2) ablating either Eye or Head features. We selected five ablation combinations, the results are displayed in Figure 12. The combinations of all features yielded

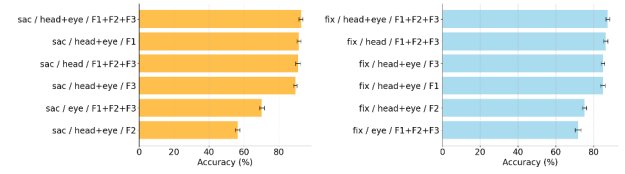


(a). For saccade classifier

(b). For fixation classifier

Figure 11: For a single saccade classifier or fixation classifier, the accuracy as a function of T across different models.

the best results for both classifiers, which justifies the decision to **use the full set of features**.



(a). For saccade classifiers

(b). For fixation classifiers

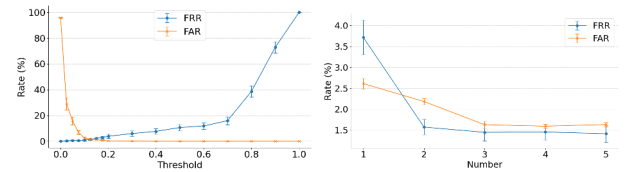
Figure 12: Ablation experiment results for the saccade and fixation classifiers with different ablation conditions. Error bars indicate standard error.

5.6 Classifiers Ensembling and Threshold Adjustment for Security

As discussed in Section 5.4, we ensemble classifiers in this section to achieve higher security. To evaluate this, we employed two commonly employed metrics [65, 67]:

False Rejection Rate (FRR): The percentage of legitimate samples incorrectly rejected.

False Acceptance Rate (FAR): The percentage of illegitimate samples incorrectly accepted.



(a). Threshold

(b). Number of classifier pairs

Figure 13: The FRR and FAR as (a) the threshold changes from 0 to 1 and (b) the number of classifier pairs for saccade and fixation classifiers changes from 1 to 5. The error bars indicate the standard error.

We ensembled the fixation classifiers with the saccade classifiers using a voting mechanism [51]. Additionally, we adjusted the positive sample threshold in binary classification to balance the FRR and FAR, as the default settings resulted in an imbalance between the two rates. Specifically, each saccade and fixation classifier outputs a predicted probability of the positive label. If the probability exceeds the set threshold, the classifier predicts a positive label; otherwise, it predicts a negative label. The final label is determined by taking a majority vote among all the predicted labels from multiple saccade and fixation classifiers.

Figure 13(a) shows how FRR and FAR vary as the threshold changes from 0 to 1 across 5 training trials with all classifiers. We selected a final threshold value of 0.125, as the FRR and FAR differ by only 0.1% at this threshold.

To optimize CoordAuth's usability, we examined the impact of varying pattern lengths (defined as the number of points). Figure 13(b) illustrates the effect of the number of classifier pairs (as indicated by the blue dashed box in Figure 9) on model performance. As most patterns contain fewer than six points, the figure covers pairs from 1 to 5. Increasing the number of pairs from 1 to 3 significantly reduces both the FAR and FRR, with relatively low values at 3 pairs (FAR = 1.6%, FRR = 1.5%). Each pair consists of saccade and fixation classifiers trained on their respective segments, so 3 pairs require at least 3 saccade and 3 fixation segments. Therefore, we recommend using patterns with **at least 4 points**, which maintain low FRR and FAR even in the event of the password collision.

5.7 Longitudinal Stability Validation

To assess the longitudinal stability of CoordAuth, we randomly selected 10 participants (6 females, 4 males, mean age = 23.9, SD = 2.6) from those in Section 5.4. Each participant additionally created two pattern passwords, following the minimum four-point requirement and entered each pattern ten times. Participants returned on Days 3 and 7 to re-enter the same passwords ten times, simulating daily mobile unlocking [21]. This yielded a total of 600 samples (2 passwords $\times$ 10 trials $\times$ 10 participants $\times$ 3 days). One participant forgot both passwords on Day 3, and those data was excluded. The remaining nine participants demonstrated 100% password recall on Days 3 and 7. A simulation showed that the binary ensemble classifier, trained on the first day's (Day 0) data with 5 trials, exhibited a slight increase in both FAR and FRR over time. FAR increased from 0.6% on Day 0 to 6.7% (SE = 2.2%) on Day 3, and to 8.2% (SE = 3.2%) on Day 7. FRR increased from 0% on Day 0 to 2.6% (SE = 1.1%) on Day 3, and to 1.4% (SE = 0.7%) on Day 7. **Despite these increases, the relatively low error rates indicate that CoordAuth remains robust over time, supporting its potential for practical use in authentication.**

5.8 Resilience Against Shoulder Surfing Attack

We conducted a study to evaluate the resilience of CoordAuth against shoulder surfing attack in real-world scenarios. We recruited 16 participants (7 males, 9 females, average age = 24.1 years, SD = 2.3) and organized them into 8 pairs. Each pair took turns assuming the role of the attacker. Participants received a compensation of \$5 each. For each trial, one participant acted as the legitimate user, while the other served as the attacker. **The legitimate user** was briefed on the experiment and seated, then asked to enter a self-created pattern password three times. **The attacker**, positioned 1 meter to the left, observed and recorded the entries on paper [67]. Afterward, the attacker used the recorded passwords to make five attack attempts on the headset. This procedure was repeated for two rounds, each with different passwords. Feedback on the success of each attack was provided. After both rounds, the roles were switched, and the process was repeated. The session concluded with a brief 5-minute interview, and the entire experiment took approximately 20 minutes.

We got a total of 160 attack attempts (16 participants $\times$ 2 patterns per participant $\times$ 5 attempts per pattern). Of these attempts, 16 successfully guessed the pattern, resulting in a success guessing rate of 10%. Among these 16 successful guesses, only one attempt successfully passed the CoordAuth behavioral biometric authentication, leading to an **Attack Success Rate (ASR)** of $1/160 = 0.6\%$.

We also compared the ASR from the previous study [67] for PIN, Pattern, and BlinkEye at an attack distance of 1m with CoordAuth. The ASRs were 18.0% (PIN), 22.5% (Pattern), 4.7% (BlinkEye), and

0.6% (CoordAuth). This shows **CoordAuth's ASR is only 3.3% of the PIN's, 2.7% of the pattern's, and 12.8% of BlinkEye's**, demonstrating CoordAuth's **robustness against shoulder surfing attacks**.

6 STUDY 2: USABILITY EVALUATION

This section evaluates CoordAuth's usability in real authentication tasks, comparing it with widely used authentication methods.

6.1 Participants and Apparatus

This IRB-approved study recruited 20 participants (12 males, 8 females) with a mean age of 22.9 (SD = 2.8). None of them participated in previous experiments. Each participant received a compensation of 15\$ for their time. We used the Meta Quest Pro Headset and the Oculus Quest Link to establish communication between the computer and the headset. We used Lenovo R9000P laptop (with an Intel Chip I9-12700) for processing the data.

6.2 Study Design

We used a two-factor within-subjects design with *technique* and *posture* as two factors. We compared the proposed method, CoordAuth with three common authentication techniques [56] (two barehanded interaction, one gaze-based interaction, see Figure 14): **Laser** [56]: 3 $\times$ 3 point grid pattern using laser-pointing with pinch confirmation, located 1.5m in front of the user. Dots were 15cm in diameter with 36cm spacing between centers.

Touch [56]: Standard numeric keypad using mid-air finger touch, positioned 0.5m in front of the user. Each square key measured 7cm in length.

Gaze: Only gaze-based 3 $\times$ 3 point grid pattern with the same setting as CoordAuth to facilitate comparison. (The ablation of CoordAuth, excluding the behavioral biometric authentication factor.)

We decided not to implement BlinkEye [67], RoomUnlock [15] and other techniques due to the lack of open-source availability for their codes and methodological details. Laser-based methods utilized the Hand Pointer Pose from the Oculus Interaction SDK, with selection confirmed by pinch gestures. Touch-based methods registered selections upon finger-tip contact with the interface.

For posture, we selected three different contextual conditions: standing, sitting, and walking postures, which consisted the majority of VR's usage case [69], facilitating the test of CoordAuth's robustness. The user interface was adjusted to face directly to participants' head position.

For each authentication technique and each posture, participants needed to first register the authentication code, then perform three unlock attempts. The registration process required 2 attempts for Laser, Touch, Gaze, and 5 attempts for CoordAuth. As decided in Section 5.4, participants were asked to input passwords with a length of at least 4 in CoordAuth, while they were asked to input passwords with a length of at least 6 in all baseline techniques. The longer code for baselines ensure adequate security and sufficient code space to resist brute-force attacks [16]. In contrast, CoordAuth's two-factor approach allows for a shorter gesture while maintaining high security, enabling fair comparison. Participants were instructed to think out different codes for different techniques they have not used beforehand. They were further instructed to avoid simple combinations like "123456", and pattern mismatches prompted a restart of registration. Registration data was used for training authentication models. In the login phase, participants received immediate feedback on authentication status after each attempt. The input data was used for testing the models.

We measured time, accuracy during the registration and authentication process for objective scales. We further measured subjective ratings for subjective scales (7-point Likert-scale): **Generalizability**: the ability to apply the authentication to different contexts and devices. **Mental Load**: the cognitive effort required to complete

the registration and authentication. **Physical Load:** the physical effort required to complete the registration and authentication. **Satisfaction:** overall user contentment with the authentication based on the effectiveness and components. **Novelty:** the degree to which the authentication is unconventional or innovative. **Security:** the vulnerability of the system to unauthorized access when observed by others. **Social Acceptability:** the appropriateness of using the authentication in public considering social norms and convenience.

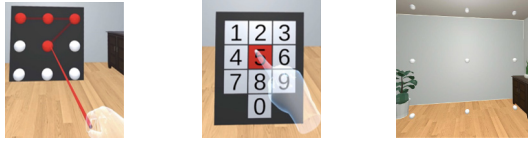


Figure 14: The illustration of authentication techniques. Left: Laser, Middle: Touch, Right: Gaze and CoordAuth.

6.3 Procedure

We introduced the experiment task to the participants, asked them to sign the user consent, calibrate the eye gaze and adjust the position. We then gave them 5 minutes to become familiarize with the apparatus and different techniques. Then participants needed to undergo two sessions for each technique and each posture: registration and login. For registration, they needed to perform corresponding attempts for sitting, standing, and walking posture each. While for login, they further needed to perform 3 consecutive attempts for each session. The order of the technique were counter-balanced for participants using Latin-square design [8]. After the experiment, participants filled a 7-point Likert-scale questionnaire for each authentication techniques following the previous design [65] and participate in the exit interview. The entire study lasted approximately 45 minutes.

6.4 Results

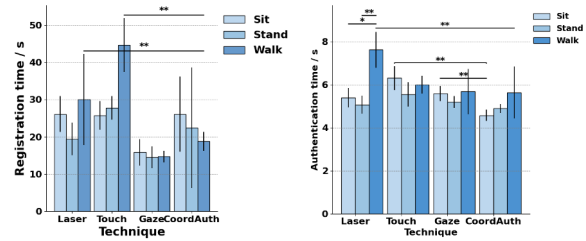
We conducted statistical analysis beginning with the Shapiro-Wilk normality test, followed by RM-ANOVA for parametric data and the corresponding non-parametric test. Post-hoc analyses were performed using the Nemenyi test for subjective ratings and the Tukey HSD test for time and error rate.

6.4.1 Time Consumption

The registration and authentication time were shown in Figure 15. The factor *techniques* had a significant effect on registration time ($F_{3,57} = 3.50, p < .05$). Post-hoc comparisons found significant difference between *Gaze* and other authentication techniques (all $p < .05$). However, there is no significant difference between *CoordAuth* (Sit: 26.1s, Stand: 22.4s, Walk: 18.8s), *Laser* (Sit: 26.1s, Stand: 19.5s, Walk: 30.1s) and *Touch* (Sit: 25.7s, Stand: 27.8s, Walk: 44.7s). The registration time in touch and laser based authentication was mainly due to the difficulty to control the pointing on the interface, resulting in extensive failure trials during registration [16]. This indicates that **CoordAuth achieves comparable registration time to commercial baselines**, even with more registration attempts.

Techniques also had a significant effect on authentication time ($F_{3,57} = 3.04, p < .05$), with post-hoc comparisons revealing significant differences between *CoordAuth* ($M = 4.6s$), *Gaze* ($M = 5.6s$) and *Touch* ($M = 6.3s$) in sitting posture, *CoordAuth* ($M = 5.6s$), *Laser* ($M = 7.6s$) in walking posture (post-hoc $p < .01$). Additionally, **posture did not have a significant effect on the registration time** ($p = .37$) **or the authentication time** ($p = .07$) for **CoordAuth, confirming its feasibility**.

CoordAuth outperformed BlinkKey [67] and RoomUnlock [15] in input and registration times, with BlinkKey requiring 9.6 seconds



(1) Registration time.

(2) Authentication time.

Figure 15: The (1) registration time and (2) authentication time of the authentication. Errorbar indicated one standard deviation. Significant differences between conditions were from post-hoc analyses.

and RoomUnlock 8.58–14.33 seconds for input, both longer than CoordAuth, while its registration time was 20 seconds shorter than BlinkKey's.

6.4.2 False Rejection Rate

Figure 16 showed the false rejection rate (FRR) of different techniques. We found a significant effect of technique on FRR ($F_{3,57} = 6.55, p < .001$). For the three knowledge-based methods, the FRR was caused by input errors. In the case of CoordAuth, the FRR was influenced by both the input errors and the false rejection of behavioral biometric authentication factor. The results reflected that, although *CoordAuth* had an additional authentication factor, the second factor never failed during participants' authentication. Additionally, posture has no significant effect on the FRR for CoordAuth, which demonstrated its robustness. **CoordAuth's FRR** (Sit: 3.2%, Stand: 3.2%, Walk: 5.4%) was **comparable to the Gaze** (Sit: 6.3%, Stand: 4.8%, Walk: 4.6%) authentication baseline, while **outperforming Laser** (17.4%) and **Touch** (15.8%) authentication in the **sitting posture**, **Touch** (19.0%) authentication in the **standing posture** and **Laser** (15.4%) and **Touch** (21.2%) in the **walking posture** (all post-hoc $p < .05$). Moreover, in the sitting posture, *CoordAuth's* FRR was also significantly lower than that of traditional pattern-based authentication methods reported in previous literature [16], highlighting the effectiveness of eye gaze. Compared to other authentication methods like BlinkKey (6.1% FRR) [67] and RoomUnlock (3.5% FRR) [15], CoordAuth achieved superior performance with a lower FRR of 3.2.

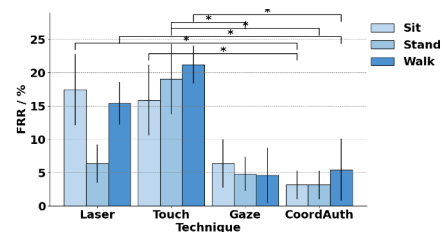


Figure 16: The FRR for different techniques. Errorbar indicated one standard deviation. Significant differences between conditions were from post-hoc analysis.

6.4.3 Subjective Feedback

CoordAuth received positive feedback as shown in Figure 17, with the factor *techniques* exhibiting significant effects across multiple dimensions. Additionally, post-hoc comparisons found significant differences between *CoordAuth* and both *Touch* and *Laser* au-

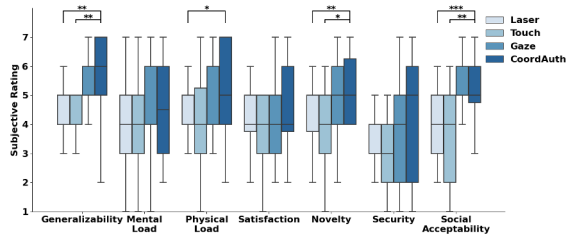


Figure 17: The subjective ratings of different dimensions (7: most positive, 1: most negative). Errorbar indicated one standard deviation. The significant differences between conditions were from post-hoc analyses.

thentication regarding generalizability, novelty and social embarrassment. **These differences highlighted CoordAuth’s superior social-acceptability, generalizability and novelty.** Participants also highlighted that CoordAuth is highly innovative and enjoyable, as mentioned by P9. P16, P17 mentioned that CoordAuth is highly efficient and agile, prompting them to authenticate fast and accurately. However, CoordAuth faced some criticisms. P7 mentioned that the overall aesthetic design of the UI could be further improved. P9 commented that entering the password using eye gaze may cause fatigue.

7 DISCUSSIONS

7.1 Feasibility of CoordAuth

Usability: In CoordAuth, we introduced a user-friendly pattern password entry authentication method in VR, leveraging natural head-eye coordination for gaze input. The gaze-based head-eye coordination input method is considered the most natural [43]. CoordAuth relies on natural head-eye interaction, requiring no hands, making it suitable for users with upper limb disabilities. The freedom from hand controllers enhances VR device portability.

Security: CoordAuth combines pattern passwords with behavioral biometrics of head and eye movements, achieving a lower FAR (see Figure 16) compared with most previous authentication technologies [67, 15]. Additionally, out of 160 attack attempts by 16 attackers, only one succeeded, outperforming PIN and pattern authentication and two-factor authentication [67] (see Section 5.8).

Reliability: Compared to applying time series classification (TSC) directly to raw data for behavioral biometric authentication [37, 31, 59], CoordAuth provides interpretability for the behavioral biometric authentication in a 2FA system. Study 1 explored unique patterns during the coordinated input of head and eye movements. The persistence of “head mover” and “non-head mover” concepts [12] in VR supports the effectiveness of CoordAuth’s biometric authentication.

Generalizability: The extracted head-eye coordination features are versatile and can be applied to other head-mounted displays (HMDs) equipped with IMUs and eye trackers of similar sampling rates and accuracy, such as AR devices [56]. Usability study explored the robustness of CoordAuth under three different postures, highlighting the algorithm’s potential in everyday social contexts within VR/AR environments [19].

7.2 Design Implications

Our design and evaluation of the head-eye coordinated authentication system revealed several important insights:

- First, head-eye coordination patterns are effectively triggered when the grid size is 40 degrees or larger. We recommend designing the interface with an grid size of at least 40 degrees

to fully exploit these coordinated motion patterns. This design choice not only maximizes input speed but also minimizes error rates by facilitating smoother head-eye interaction.

- Second, the authentication algorithm should be optimized to leverage head-eye coordination by selecting displacement and rotation, velocity, and frequency spectrum features, and exploring their combinations.
- Lastly, integrating gaze-based input into established authentication methods can enhance both user experience and efficiency. Future research could focus on integrating gaze-based inputs on traditional authentication or interactions, such as using gaze tracking to guide users’ selections.

8 ETHICAL CONSIDERATIONS

We adhered to Menlo report and Belmont report for conducting the research. All studies were approved by Tsinghua University Science and Technology Ethics Committee, with protocol number THU-03-2024-0005. Participants signed consents before participating the study. We properly compensated the participants and provided them options to quit the experiment.

9 LIMITATION AND FUTURE WORK

We acknowledged the limitations in our work, which we also regarded as future directions. First, in our study, we exclusively utilized the Quest Pro VR headset for all experiments. The Quest Pro’s signal sampling rate is 50Hz. The balance between insufficient biometric feature capture at lower sampling rates and the potential redundancy and resource demands of higher sampling rates remains an open question in optimizing authentication performance. Additionally, our participants consisted solely of college students aged 18 to 30. We recognize the need for future experiments involving a more diverse group of participants.

Second, in the simulation study we investigated the stability of CoordAuth only over a 7-day period. Future research should explore the stability of CoordAuth with longer periods. In the usability study, we only validated CoordAuth for usability in sitting, standing, and walking postures [19]. Future research should focus on its performance in a wider range of daily mobile interaction postures and its robustness to noise.

Last, while CoordAuth currently employs a 3×3 grid UI, the potential performance improvement of CoordAuth with a 4×4 or other UI designs is a topic worthy of discussion. Additionally, the relatively simple pattern used in Study 1 is a limitation that should be considered. Furthermore, given that CoordAuth involves gaze-based biometric features, there is potential for implicit authentication if active input is replaced with stimuli. For instance, integrating gaze-based biometric authentication into gaze calibration might be a promising direction for future research.

10 CONCLUSION

This paper presents CoordAuth, a hands-2FA for VR, utilizing head-eye coordination. CoordAuth integrates gaze-based interaction for entering 3×3 pattern passwords as knowledge-based authentication, while also employing head and eye movement features for behavioral biometric authentication. We conducted a user study to optimize the input interface size. Through a voting mechanism and novel feature extraction techniques, CoordAuth demonstrates both strong security and long-term stability. Further studies confirm CoordAuth’s high usability and robustness against shoulder-surfing attacks, establishing it as a promising authentication solution.

ACKNOWLEDGEMENTS

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